

Climate 5000 Series - Gen 4

9K BTU Single Zone Regular System

Wall Mounted IDU

Engineering
Submittal
Sheet



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Overview and Project Information



Project name

Location

Architect

Engineer

General contractor

Reference #

9K Btu Single Zone Regular System - Wall Mounted IDU

Indoor part number 8-733-962-672

Indoor model number BMS500-AAU009-1AHWXD

Outdoor part number 8-733-962-684

Outdoor model number BMS500-AAS009-1CSXRD

Efficiency Data AHRI 210/240 2023

SEER2 25.8

EER2 14.9

HSPF2 Region IV 12.5

HSPF2 Region V 9.2

COP(47F) 3.90

COP (5F) 2.36

Performance Data AHRI 210/240 2023

Capacity Range 2200~13000

Rated Capacity 9000

Current 4.5

Input Power 604

Capacity Range 4100~13800

Rated Capacity (47F) 10000

Rated Capacity (5F) 8000

Current 3.3

Input Power 750

Electrical Data

Power supply (V / Ø / Hz) 208-230 / 1 / 60

Min. circuit ampacity indoor/outdoor (A) 3.0 / 12.0

Max. fuse indoor/outdoor (A) 15.0 / 15.0

Compressor Data

Type: Rotary

Refrigerant oil /oil charge (ml) VG74 / 310

Input (W) 325 / 826

Rated current (RLA/A) 2.40 / 5.65

Operation Range

Cooling (°F / °C) -13~-122 / -25~-50

Heating (°F / °C) -13~-75 / -25~-24

Cooling (°F / °C) 60~90 / 16~32

Heating (°F / °C) 32~86 / 0~30

Piping

Refrigerant precharge (oz/kg) 32.45 / 0.92

Length for precharge (ft/m) 24.6 / 7.5

Liquid side (in/mm) 1/4" / Ø6.35

Gas side (in/mm) 3/8" / Ø9.52

Additional Charge of Refrigerant (oz/ft)/(g/m) 0.16 / 15

Max. refrigerant pipe length (ft/m) 82 / 25

Max. difference in height (ft/m) 49.2 / 15

System Data

Refrigerant type R454B

Design pressure (PSIG) 550 / 340

Dehumidification (pts/hr) 1.27

Fan motor model ZKFP-20-8-6-7

Fan motor qty (ea) 1

Fan motor input (W) 50.0

Fan speed (RPM) 1100 / 850 / 700

Air flow, Turbo / H / M / L / Si (CFM) 441.4 / 323.7 / 229.6 / 176.6 / 147.2

Approx. Air Throw Range ft (m) 23.0 / 7

Noise level, Turbo / H / M / L / Si (dB(A)) 47.5 / 43 / 31 / 18 / 17

Net weight (lbs / kg) 23.37 / 10.6

Gross weight (lbs / kg) 29.98 / 13.6

Fan motor model ZKFN-34-10-1L

Fan motor qty (ea) 1

Fan motor input (W) 50

Fan motor speed, H / M / L (RPM) 800 / 700 / 600

Air flow, Max (CFM) 1235.3

Noise level, Max (dB(A)) 52.5

Net weight (lbs / kg) 62.17 / 28.2

Gross weight (lbs / kg) 67.68 / 30.7

Accessories

BMS-WT2-XXD (8733962707) Wired Wall Thermostat

BMS-DRY-XXD (8733969263) 24V Interface

BMS-PK1-7X1 (8733951010) 25ft Lineset

Standard Parts

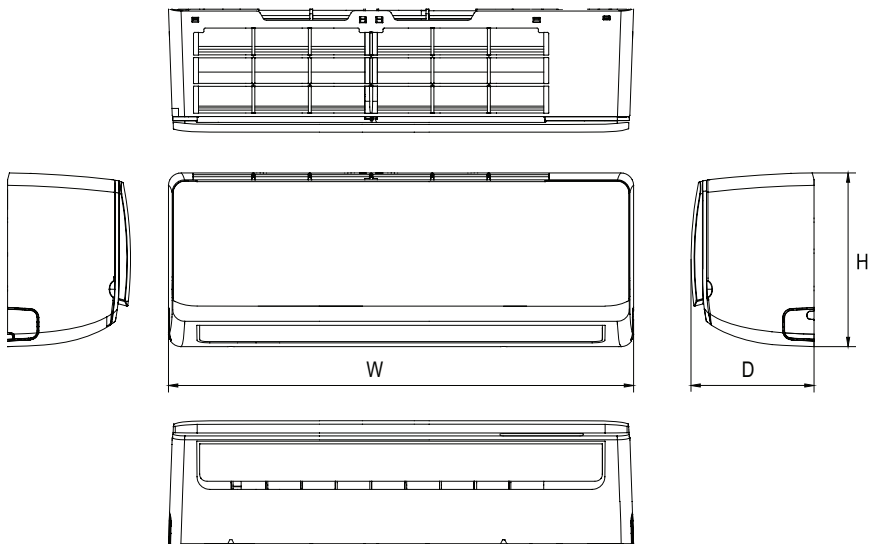
8-733-969-088 Remote Control

Climate 5000 Series - Gen 4

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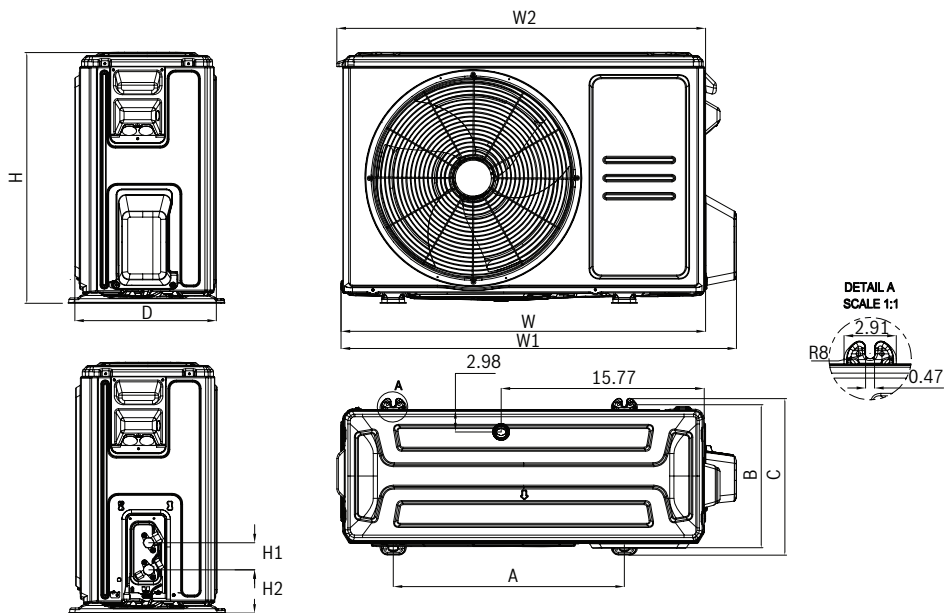


Indoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)
BMS500-AAU009-1AHWXD	31.42	9.21	11.85

Outdoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)	W1 (in)	W2 (in)	H1 (in)	H2 (in)	A (in)	B (in)	C (in)
BMS500-AAS009-1CSXRD	30.12	11.93	21.85	32.9	30.87	2.36	3.67	17.8	11.26	12.36

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Cooling Performance - Indoor Dry Bulb Temperature at 441 CFM																					
ODU DB(°F)	IDU WB(°F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)				
	IDU DB(°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-13°F (-25.0°C)	TC	10480	10476	10625	10669	10963	10935	10926	10919	10918	10909	11262	11248	11237	11235	11225	11879	11876	11851	11847	11822
	S/T	0.80	0.91	1.00	1.00	1.00	0.65	0.75	0.85	0.86	0.97	0.55	0.65	0.74	0.76	0.86	0.39	0.47	0.56	0.57	0.67
	PI	0.75	0.75	0.75	0.74	0.74	0.74	0.74	0.74	0.74	0.74	0.74	0.74	0.74	0.74	0.74	0.75	0.75	0.75	0.75	0.75
-4°F (-20.0°C)	TC	11287	11281	11442	11490	11806	11776	11766	11759	11758	11748	12128	12113	12101	12100	12088	12793	12789	12763	12758	12731
	S/T	0.80	0.92	1.00	1.00	1.00	0.65	0.76	0.85	0.87	0.98	0.55	0.65	0.74	0.76	0.86	0.39	0.48	0.56	0.57	0.67
	PI	0.77	0.77	0.77	0.77	0.77	0.77	0.77	0.77	0.77	0.77	0.77	0.77	0.77	0.77	0.77	0.77	0.77	0.77	0.77	0.77
0°F (-17.8°C)	TC	11661	11654	11870	11920	12247	12161	12148	12138	12137	12259	12519	12501	12485	12484	12478	13192	13189	13188	13190	13198
	S/T	0.81	0.93	1.00	1.00	1.00	0.66	0.77	0.86	0.88	1.00	0.55	0.66	0.75	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.79	0.79	0.79	0.79	0.79
5°F (-15.0°C)	TC	11995	12037	12312	12369	12750	12558	12561	12562	12575	12762	12960	12963	12966	12966	12983	13720	13716	13723	13724	13730
	S/T	0.82	0.94	1.00	1.00	1.00	0.66	0.78	0.87	0.88	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.59	0.69
	PI	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.80	0.80	0.81	0.81	0.81	0.81	0.81
14°F (-10.0°C)	TC	12254	12248	12498	12556	12946	12862	12851	12842	12841	12961	13303	13286	13273	13271	13257	14145	14141	14114	14109	14079
	S/T	0.81	0.93	1.00	1.00	1.00	0.65	0.76	0.86	0.87	1.00	0.55	0.66	0.75	0.76	0.87	0.39	0.48	0.56	0.57	0.67
	PI	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79
17°F (-8.3°C)	TC	12311	12351	12636	12698	13110	12924	12929	12931	12942	13125	13365	13370	13375	13376	13392	14214	14210	14208	14210	14222
	S/T	0.82	0.93	1.00	1.00	1.00	0.66	0.77	0.87	0.88	1.00	0.55	0.66	0.76	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79
23°F (-5.0°C)	TC	12818	12873	13206	13272	13709	13467	13471	13473	13491	13724	13932	13937	13940	13941	13963	14814	14809	14820	14822	14830
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.89	0.39	0.48	0.57	0.59	0.69
	PI	0.79	0.79	0.79	0.79	0.80	0.80	0.80	0.80	0.80	0.80	0.80	0.80	0.80	0.80	0.80	0.82	0.82	0.82	0.82	0.82
35°F (1.7°C)	TC	13961	13961	14044	14083	14537	14614	14613	14614	14614	14653	15071	15070	15071	15071	15073	15920	15916	15915	15914	15916
	S/T	0.78	0.88	0.97	0.99	1.00	0.63	0.73	0.82	0.83	0.93	0.54	0.63	0.72	0.73	0.83	0.39	0.46	0.54	0.56	0.65
	PI	0.75	0.75	0.76	0.76	0.79	0.80	0.80	0.80	0.80	0.80	0.83	0.83	0.83	0.83	0.83	0.90	0.90	0.90	0.90	0.90
41°F (5.0°C)	TC	12418	12418	12517	12604	12976	12995	12995	12996	12996	13041	13401	13400	13401	13401	13402	14153	14149	14148	14148	14149
	S/T	0.78	0.89	0.98	1.00	1.00	0.64	0.74	0.83	0.84	0.94	0.54	0.64	0.72	0.74	0.84	0.39	0.47	0.55	0.56	0.66
	PI	0.58	0.58	0.59	0.59	0.62	0.62	0.62	0.62	0.62	0.62	0.64	0.64	0.64	0.64	0.65	0.70	0.70	0.70	0.70	0.70
47°F (8.3°C)	TC	10873	10880	11026	11076	11403	11377	11377	11377	11377	11430	11731	11730	11730	11730	11730	12387	12384	12383	12383	12384
	S/T	0.79	0.90	1.00	1.00	1.00	0.64	0.75	0.84	0.85	0.96	0.54	0.65	0.73	0.75	0.85	0.39	0.47	0.55	0.57	0.67
	PI	0.41	0.41	0.42	0.42	0.44	0.44	0.44	0.44	0.44	0.44	0.46	0.46	0.46	0.46	0.46	0.50	0.50	0.50	0.50	0.50
50°F (10.0°C)	TC	10658	10952	11321	11387	11828	11260	11266	11344	11380	11840	11699	11715	11721	11721	11834	12506	12539	12571	12575	12591
	S/T	0.90	1.00	1.00	1.00	1.00	0.71	0.85	0.96	0.99	1.00	0.59	0.72	0.83	0.85	0.99	0.40	0.50	0.60	0.62	0.74
	PI	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.47
59°F (15.0°C)	TC	10598	10970	11359	11428	11893	11213	11228	11356	11440	11905	11676	11687	11691	11709	11915	12537	12562	12587	12590	12601
	S/T	0.92	1.00	1.00	1.00	1.00	0.73	0.86	1.00	1.00	1.00	0.60	0.73	0.85	0.86	1.00	0.40	0.51	0.61	0.63	0.76
	PI	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.51	0.51	0.51	0.51	0.51
68°F (20.0°C)	TC	10533	10982	11390	11462	11948	11150	11176	11402	11474	11960	11633	11641	11662	11672	11970	12534	12557	12577	12579	12586
	S/T	0.95	1.00	1.00	1.00	1.00	0.74	0.89	1.00	1.00	1.00	0.61	0.75	0.87	0.89	1.00	0.40	0.51	0.63	0.65	0.78
	PI	0.53	0.53	0.54	0.54	0.54	0.53	0.53	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.55	0.55	0.54	0.54	0.54
77°F (25.0°C)	TC	10459	10963	11392	11468	11980	11088	11123	11404	11481	11992	11590	11603	11633	11646	12002	12518	12556	12585	12588	12600
	S/T	0.97	1.00	1.00	1.00	1.00	0.75	0.90	1.00	1.00	1.00	0.61	0.76	0.88	0.90	1.00	0.40	0.52	0.63	0.65	0.79
	PI	0.56	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.58	0.58	0.58	0.58	0.58
82°F (27.8°C)	TC	10592	11133	11582	11661	12195	11225	11268	11592	11674	12207	11747	11762	11798	11813	12218	12701	12749	12782	12785	12799
	S/T	0.99	1.00	1.00	1.00	1.00	0.76	0.91	1.00	1.00	1.00	0.62	0.77	0.89	0.92	1.00	0.40	0.52	0.64	0.66	0.80
	PI	0.63	0.63	0.63	0.64	0.64	0.63	0.63	0.63	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64
86°F (30.0°C)	TC	10956	11503	11973	12057	12618	11554	11607	11986	12070	12631	12101	12115	12160	12177	12642	13101	13151	13183	13186	13198
	S/T	1.00	1.00	1.00	1.00	1.00	0.77	0.93	1.00	1.00	1.00	0.62	0.78	0.91	0.93	1.00	0.40	0.52	0.64	0.66	0.81
	PI	0.74	0.74	0.74	0.74	0.75	0.74	0.74	0.74	0.74	0.75	0.75	0.74	0.74	0.75	0.75	0.75	0.75	0.75	0.75	0.75
95°F (35.0°C)	TC	11694	11757	12137	12209	12692	12428	12436	12440	12454	12708	12962	12964	12972	12974	12995	13967	13962	13964	13968	13987
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.58	0.68
	PI	0.94	0.94	0.96	0.96	0.98	0.97	0.97	0.97	0.97	0.98	0.99	0.99	0.99	0.99	0.99	1.03	1.03	1.03	1.03	1.03
104°F (40.0°C)	TC	10688	10786	11192	11259	11708	11361	11368	11388	11400	11723	11846	11851	1185							

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Heating Performance - Indoor Dry Bulb Temperature at 441 CFM										
Outdoor DB(°F)	TCI Total Heating Capacity(BTU/hr)					PI Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-13°F(-25.0°C)	5949	5830	5800	5777	5725	0.99	1.00	1.00	1.00	1.00
-4°F(-20.0°C)	7772	7616	7578	7548	7480	1.09	1.10	1.10	1.10	1.11
0°F(-17.8°C)	7899	7741	7702	7672	7603	1.05	1.06	1.06	1.06	1.07
5°F(-15.0°C)	8058	7897	7858	7826	7756	1.00	1.01	1.01	1.01	1.02
14°F(-10.0°C)	9205	9021	8976	8940	8860	1.11	1.11	1.12	1.12	1.12
17°F(-8.3°C)	9588	9396	9349	9311	9228	1.14	1.15	1.15	1.15	1.16
22°F(-5.6°C)	9787	9591	9543	9505	9419	1.14	1.15	1.16	1.16	1.16
27°F(-2.8°C)	9986	9786	9737	9699	9611	1.15	1.16	1.16	1.16	1.17
32°F(0.0°C)	10185	9982	9932	9892	9803	1.15	1.16	1.17	1.17	1.17
35°F(1.7°C)	10305	10099	10048	10008	9918	1.16	1.17	1.17	1.17	1.18
37°F(2.8°C)	10939	10721	10667	10624	10529	1.15	1.16	1.16	1.17	1.17
42°F(5.6°C)	12526	12275	12214	12165	12055	1.14	1.15	1.15	1.15	1.15
44.6°F(7.0°C)	13160	12897	12833	12781	12666	1.13	1.14	1.14	1.14	1.15
47°F(8.3°C)	14112	13830	13761	13706	13582	1.12	1.13	1.13	1.14	1.14
52°F(11.1°C)	13957	13678	13610	13555	13433	1.02	1.03	1.03	1.04	1.04
57°F(13.9°C)	13802	13526	13459	13405	13284	0.93	0.93	0.94	0.94	0.94
59°F(15.0°C)	13740	13466	13398	13345	13225	0.89	0.89	0.90	0.90	0.90
62°F(16.7°C)	13647	13374	13308	13254	13135	0.83	0.83	0.84	0.84	0.84
64.4°F(18.0°C)	13573	13302	13235	13182	13064	0.78	0.79	0.79	0.79	0.79
69.8°F(21.0°C)	13406	13138	13072	13020	12903	0.68	0.68	0.68	0.68	0.69
75.2°F(24.0°C)	13239	12974	12909	12857	12742	0.57	0.57	0.58	0.58	0.58

Bolded values were tested. All other values are simulated and for reference only.